

Practice Questions: Queueing Theory

Solution

Question 1

[Simple M/M/1] Utopia city has been struck by a crime wave. Alarmed by the increasing number of bank robberies and concerned about their effect on bank customers, the Banking Upper Management Society (BUMS) adopts the following policies:

- i. The bank robbers reserve a teller's window for the exclusive use.
- ii. To conserve space, only a lone bandit may commit bank robberies.
- iii. If two or more robberies occur simultaneously, the robbers are served on a first-come, first-served basis.

The Bank Robbers Federation (BARF) engages you as a consultant. Your job is to determine if the present arrangements with the banks are adequate.

You have been given the following information - At each bank:

- Robbers arrive at random between 9.00am and 3.00pm; the average arrival rate is *five robbers per hour*.
- Teller service time is *exponential* with an average of *6 minutes* (for the robbers' teller).
- The M/M/1 model seems to apply.

Determine the following:

- a. The average time a robber must queue for service (a robbery).
- b. The average time required for a robbery (queueing time plus service time).
- c. The probability that the robbers' teller is busy.
- d. The average number of robbers in the bank.
- e. The probability of finding three or more robbers in the bank at the same time.
- f. The probability of a robbery requiring more than half an hour.
- g. If the average robber arrival rate increases to *9 per hour* next year (an 80% increase), what will be the *average robbery time* then (queue for service plus the actual robbery)?

$$\lambda = \frac{5}{60} = \frac{1}{12} \text{ robbers / minute}$$

By (ii) $E[s]$ is 6 minutes

$$\rho = \lambda E[s] = \frac{6}{12} = 0.5$$

(a) average time a robber must queue for service, $E[w]$

$$\begin{aligned} E[w] &= \frac{\rho E[s]}{1 - \rho} \\ &= \frac{0.5 * 6}{0.5} = 6 \text{ min} \end{aligned}$$

(b) average time required for a robbery

$$\begin{aligned} E[r] &= \frac{E[s]}{1 - \rho} \\ &= 12 \text{ min} \end{aligned}$$

(c) Probability that the robber's teller is busy
 $= \rho = 0.5$

(d) Average number of robbers in the bank:

$$E[n] = \frac{\rho}{1 - \rho} = 1$$

Alternatively, by Little's formulae,

$$E[n] = \lambda E[r] = 1$$

(e) $\rho^n = 0.5^3 = 0.125$

$$\begin{aligned} \text{(f) } p[r > 30] &= e^{-t/E[r]} & p[r > t] &= e^{-t/E[r]} \\ &= e^{-30/12} \\ &= 0.082085 \end{aligned}$$

The answer reduces to understanding the exponential function's cumulative distribution function (CDF). For a random variable (like the response time), the cumulative distribution function is the probability that X (response time of a job, R) will take a value less than or equal to x (less than 30 minutes):

$$F_X(x) = P(X \leq x) \text{ or } F_R(30 \text{ mins}) = P(R \leq 30 \text{ mins})$$

For M/M/1, the response time is exponentially distributed and has a mean value of

$$E[r] = \frac{1}{\mu(1-\rho)}$$

For an exponential distribution with average $E[r]$, the cumulative distribution function(CDF) of the response time would be

$$F(r) = 1 - e^{-r \frac{1}{E[r]}} = 1 - e^{-r\mu(1-\rho)}$$

As such, the probability that the response time is less or equal to r_q (i.e. 30 minutes) would be $1 - e^{-\frac{r_q}{E[r]}}$ (left-hand side on the CDF graph).

In conclusion, the probability that the response time is more than r_q (i.e. 30 minutes) would be $1 - \left(1 - e^{-\frac{r_q}{E[r]}}\right)$ (right-hand side on the CDF graph).

$$p(r < r_q) = e^{-r_q/E[r]}$$

(g) $\rho = \lambda E[s]$

$$= \frac{9}{60} * 6 = 0.9$$

$$\begin{aligned} E[r] &= \frac{E[s]}{1-\rho} \\ &= \frac{6}{1-0.9} = 60 \text{ min} \end{aligned}$$

Question 2

1. [Application of M/M/1 to verify Performance Issue] At a computer consulting company, programmers use an on-line terminal to the company's central computer system to debug their programs. The company operates *24-hour a day*, and *48 programmers* (on the average) need to use the terminal once a day. The programmers arrive at the terminal at random (assume Poisson arrivals). The time each programmer spends at the terminal is *exponential* and *averages 20 minutes*.

Each day the operations manager receives several complaints from the programming staff that the waits experienced at the terminal are too long and that productivity is being lost. The programmers argue that more terminals are needed. Management is concerned that the terminal is being utilized only **two-thirds** of the time, and thus argues that adding terminals seems wasteful.

Use queueing theory to analyze the situation at this company and determine if indeed the additional terminals are warranted.

$$\lambda = \frac{48}{24} = 2 \text{ prog/hour}, E[s] = 20 \text{ minutes} = \frac{1}{3} \text{ hours} \Rightarrow \mu = \frac{1}{E[s]} = 3 \text{ prog/hour}$$

$$\rho = \frac{\lambda}{m\mu} = \frac{2}{3} \Rightarrow m = 1, \text{ currently the firm has 1 terminal.}$$

$$E[w] = \frac{\rho E[s]}{1 - \rho} = \frac{(\frac{2}{3})(\frac{1}{3})}{1 - \frac{2}{3}} = \frac{2}{3} \text{ hour} = 40 \text{ minutes}$$

$$E[r] = E[w] + E[s] = \frac{2}{3} + \frac{1}{3} = 1 \text{ hour}$$

$$90^{\text{th}} \text{ percentile queueing time is } \frac{E[w]}{\rho} \ln(10\rho) = \frac{\frac{2}{3}}{\frac{2}{3}} \ln(10 \cdot \frac{2}{3}) = 1.9 \text{ hours} = \mathbf{113.8 \text{ minutes}}$$

10% of the programmers (≈ 5 per day) have to wait almost two hours for a free terminal.

$$\text{Mean number of programmers in the queue, } E[n_q] = \lambda E[w] = 2(\frac{2}{3}) = \frac{4}{3} \text{ prog/hour}$$

$$\text{Mean number of programmers in the system, } E[n] = \lambda E[r] = 2(1) = 2 \text{ prog/hour}$$

Our analysis confirms that the single terminal is not sufficient for handling the programmers' need without causing excessive waits.

Question 3

[Application of M/M/1 to evaluate different computer configurations] After considering the preceding analysis (question 2), the company decided to purchase additional computer terminals.

- How many terminals should be purchased to keep the 90th percentile waiting times below ten minutes?
- Should these terminals be kept in a *central location*, or should they be located throughout the building (*decentralized*)?

You can ignore the time it takes programmers to walk to the terminals.

- How many terminals should be purchased to keep the 90 percentile waiting times below ten minutes?

M/M/2

$$\rho = \frac{2}{2 \times 3} = \frac{1}{3}$$

$$P_0 = \left\{ 1 + \frac{(2 \times \frac{1}{3})^2}{2!(1 - \frac{1}{3})} + \frac{2 \times \frac{1}{3}}{1!} \right\}^{-1} = \frac{1}{2}$$

$$P(\text{Queueing}) = \frac{(2 \times \frac{1}{3})^2}{2!(1 - \frac{1}{3})} \left(\frac{1}{2} \right) = \frac{1}{6}$$

$$E[n_q] = \frac{(\frac{1}{3})(\frac{1}{6})}{1 - \frac{1}{3}} = \frac{1}{12} \text{ programmers}$$

$$E[w] = \frac{\frac{1}{12}}{2} = \frac{1}{24} \text{ hours} = 2.5 \text{ minutes}$$

$$90\text{-percentile of the waiting time: } \frac{\frac{1}{24}}{\frac{1}{6}} \ln(10 \times \frac{1}{6}) = 0.13 \text{ hour} = 7.7 \text{ minutes}$$

Two terminals

- Should these terminals be kept in a *central location*, or should they be located throughout the building (*decentralized*)?

2 M/M/1, for each M/M1

$$\rho = \frac{1}{3}$$

$$E[w] = \frac{\rho E[s]}{1 - \rho} = \frac{(\frac{1}{3})(\frac{1}{3})}{1 - \frac{1}{3}} = \frac{1}{6} \text{ hours} = 10 \text{ minutes}$$

$$90\text{-percentile of the waiting time: } \frac{\frac{1}{6}}{\frac{1}{3}} \ln(10 \times \frac{1}{3}) = 0.6 \text{ hours} = 36.1 \text{ minutes}$$

Thus, the centralized one is better.

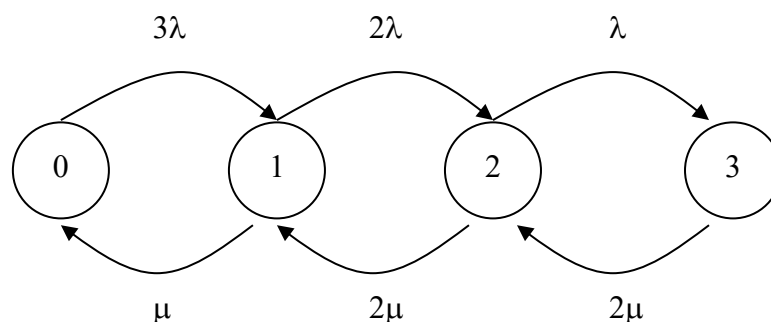
Question 4

[Formulation of solution based on first principles] Consider a queueing network with two service facilities connected in a *cyclic* fashion (input to output, input to output). One service facility is a M/M/2 queueing system. Each server operates at a rate of μ . The other service facility consists of three servers in parallel with no queue (M/M/3). The service rate of each of these servers is λ . Let n be the number of customers in the M/M/2 queueing system. The total number of customers in this closed system is 3. Assume Markovian statistics.

In terms of applications the M/M/2 system could be a dual processor unit and the M/M/3/3 system could be the number of terminals. Customers arrive and submit jobs.

- Draw and label the state transition diagram for $n=0, 1, 2, 3$.
- Write out expressions for p_n in terms of p_0, λ, μ , and n up to $n=3$.
- Let $\lambda = 1.0$ and $\mu = 1.0$. Calculate the average time it takes a customer to circulate around the system (through both queueing systems).

State transition diagram



b) Job flow balance

$$3\lambda P_0 = \mu P_1$$

$$\therefore P_1 = \frac{3\lambda}{\mu} P_0$$

$$(\mu + 2\lambda)P_1 = 3\lambda P_0 + 2\mu P_2$$

$$\therefore P_2 = \frac{3\lambda^2}{\mu^2} P_0$$

$$\lambda P_2 = 2\mu P_3$$

$$\therefore P_3 = \frac{3\lambda^3}{2\mu^3} P_0$$

c) If $\lambda = 1.0$, $\mu = 1.0$, then $P_1 = 3P_0$, $P_2 = 3P_0$, $P_3 = 1.5P_0$

Since $P_0 + P_1 + P_2 + P_3 = 1$ then

$$P_0 = \frac{1}{8.5} = 0.1176 \quad \text{Therefore, } P_1 = 0.3528, P_2 = 0.3528, \text{ and } P_3 = 0.1764$$

The response time for M/M/2 subsystem

$$E[n] = P_1 + 2P_2 + 3P_3 = 1.5876$$

$$\text{Expected arrival rate} = 3\lambda P_0 + 2\lambda P_1 + \lambda P_2 = 1.4112$$

$$\text{Expected response time} = 1.5876 / 1.4112 = \mathbf{1.125}$$

The M/M/3/3 terminal subsystem can be modeled as a delay center, therefore the expected response time is $1/\lambda = \mathbf{1}$

Thus, the total time it takes for a job (on average) to complete 1 cycle is **2.125**.

Question 5

[Model Selection and Modification Analysis] A *quad-core* database server receives *three transactions per second* and each core processes *0.8 transactions per second*. Based on the accounting log, transaction arrival is Poisson and processor core service time is exponentially distributed.

- a. What is an appropriate queueing model for studying this system? Justify your answer.
- b. Compute the *mean waiting time* and *mean response time of a transaction*.
- c. Based on the mean waiting time and mean response time computed in (b), the SLA of *two seconds* for these two metrics is not met. The system manager is considering two options: (i) *upgrade to a faster quad-core server*, or (ii) *buy two faster servers, with each server consisting of a dual-core processor*. In both (i) and (ii) the speed of each core is increased by 50%. Which option meets the SLA and what option will you recommend? Justify your answer.

[Hint: To simplify your computation, the probability that an arriving transaction must wait in the queue, ξ , for the current system, option 1 and option 2, are 0.87, 0.32 and 0.48, respectively.]

- a. Since the quad-core shares memory, has a single queue and cores, homogeneous and Poisson transaction arrivals and exponential service time, the processor can be modeled as an M/M/4 system.
- b. The means as in the spreadsheet are mean waiting time = 4.35 s and mean response time = 5.58 s.
- c. More marks are allocated in this section because students must identify the models for each of the options.

Option 1 remains to be a M/M/4 QN but with the speed of a cores increasing from 0.8 to 1.2 transactions per second. Mean response time = 1.011s

Option 2 – models as two separate M/M/2 QN, each with its own queue. Core speed is increased from 0.8 to 1.2 transactions per second. Mean response time = 1.367s

The recomputed mean waiting time and mean response time are all within the SLA of 2 seconds – meets the SLA.

But which option to choose? Students answer should relate the size and variation in size of transactions in justifying the option.